

scripts/generate-coverage-ledger.sh — User Guide

Last verified: 2026-05-16 (constitution-compliance plan Phase 7) **Inheritance:** HelixConstitution §11.4.25 (Full-Automation-Coverage Mandate) + §11.4.18 (script docs)

Overview

Generates the canonical docs/coverage-ledger.yaml — Lava's machine-readable coverage matrix per HelixConstitution §11.4.25.

For every feature module, core module, app, lava-api-go service, and owned-by-us Submodule, the generator emits one row recording:

- The path (e.g. feature/onboarding, core/auth, lava-api-go, submodules/containers)
- The kind (feature | core | app | api | submodule)
- Test counts (unit, integration) discovered on disk
- Challenges that target the row (matched by import // covers-feature: marker / filename keyword / heuristic broad-flow mapping — same matching rules as scripts/check-challenge-coverage.sh)
- Up to 3 sample test file basenames
- The six §11.4.25 invariants, each with status pass | gap | n/a | "waiver: <reason>" | "pass: <reason>"
- An overall row status (covered | partial | gap)

The six §11.4.25 invariants

Invariant	Mechanical inference	Notes
anti_bluff	pass if challenge_count > 0; else gap	Challenges carry §6.AB falsifiability rehearsals
working_capability	pass if challenge_count > 0; else gap	Challenge tests are the E2E gate per §6.AE Requires operator-or-reviewer override via waiver
documented_promise	Always defaults to gap	— typically points at a CONSTITUTION.md / README.md / spec
no_open_bugs	Always defaults to gap	Requires operator-or-reviewer override via waiver — typically references

Invariant	Mechanical inference	Notes
		closure logs under <code>.lava-ci-evidence/crashlytics-resolved/</code> per §6.AD.3 equivalence
documented	pass if any of {README.md, CLAUDE.md, AGENTS.md, CONSTITUTION.md} present in the path; else gap	Per-scope §6.AD pointer-block makes this pass for any path under <code>feature/</code> , <code>core/</code> , <code>app/</code> , <code>lava-api-go/</code> , <code>submodules/</code> (after waivers)
four_layer	pass if <code>unit_test_count > 0</code> AND <code>challenge_count > 0</code> ; else gap	§1 four-layer floor: pre-build + post-build + runtime + paired mutation

Status computation

A row's overall status is derived from the six invariant values:

pass count	gap count	Status
6	0	covered
3+	1+	partial
<3	1+	gap

`waiver:*` and `n/a` invariants count as neither pass nor gap. A row that has 3+ pass + 0 gap → covered.

Waiver mechanism

The generator default for invariants 3, 4 is gap because they cannot be mechanically inferred. To override, hand-edit `docs/coverage-ledger/waivers.yaml`:

```
waivers:
- path: feature/onboarding
  invariants:
    documented_promise: "pass: feature documented in docs/UX-
      FLOWS.md + Onboarding screen content"
    no_open_bugs: "pass: 2026-05-15 audit — no open Crashlytics
      issues per .lava-ci-evidence/crashlytics-resolved/"
```

Waiver values can be:

- `"pass: <evidence>"` — operator-or-reviewer affirms pass with positive evidence
- `"waiver: <reason>"` — explicit waiver with tracked-item reference
- `"n/a"` — invariant does not apply to this row

The waiver file is hand-edited + version-controlled. Per §6.J anti-bluff, every "pass:" waiver SHOULD cite a concrete evidence trail (closure log, commit SHA, doc path).

Usage

```
# Regenerate the ledger + print gap summary
bash scripts/generate-coverage-ledger.sh

# Emit YAML to stdout instead of writing the file
bash scripts/generate-coverage-ledger.sh --stdout

# Write silently (no stdout summary)
bash scripts/generate-coverage-ledger.sh --quiet
```

Determinism

Same git tree → same generated row content. The only varying header field is `metadata.generated_at` (UTC timestamp). The verifier (`scripts/check-coverage-ledger.sh`) strips metadata + `generated_at` before diffing so cosmetic timestamp diffs do not fail the freshness check.

Companion files

- `docs/coverage-ledger.yaml` — the generated ledger (committed)
- `docs/coverage-ledger.waivers.yaml` — hand-edited per-row waiver overrides (committed)
- `scripts/check-coverage-ledger.sh` — verifier (asserts presence + freshness + row coverage)
- `tests/check-constitution/test_coverage_ledger.sh` — hermetic falsifiability rehearsals (6 fixtures)

Cross-references

- HelixConstitution Constitution.md §11.4.25 (the mandate)
- `docs/plans/2026-05-15-constitution-compliance.md` Phase 7
- `docs/helix-constitution-gates.md` CM-COVERAGE-LEDGER row
- `scripts/verify-all-constitution-rules.sh` (wraps the verifier in advisory mode per §11.4.32)
- `Lava CLAUDE.md` §6.AD-debt (full-automation-coverage tracker integration)

Inheritance + classification

- Inheritance: HelixConstitution §11.4.25 (the mandate); §11.4.18 (script docs); §6.J/§6.L (anti-bluff posture of the generator)
- Classification: project-specific (the Lava module list is project-specific; the §11.4.25 ledger mandate is universal per HelixConstitution)